



**McKean-Vlasov model**

- Simple model: consider

$$dX_t^i = \sum_{j=1}^N a(X_t^i - X_t^j) dt + \sigma dW_t^i$$

- What is the behaviour depending on  $a$ ,  $\sigma$ ,  $N$ ?

- Can be solved explicitly,  $Y_t^i = X_t^i - \frac{1}{N} \sum_i X_t^i$  is an Ornstein-Uhlenbeck process

$$\bullet \, dY_t^i = aNY_t^i dt + \sigma \left( dW_t^i - \frac{1}{N} \sum_i dW_t^i \right)$$

•  $dY_t^i = aNY_t^i dt + \sigma\sqrt{1-1/N}dB_t$ , consider  $a = c/N$

- For  $c < 0$ ,  $Y_t^i \rightarrow N(0, \tilde{\sigma}(N, c))$  in law as  $t \rightarrow +\infty$  and

$$\frac{1}{N} \sum_i X_t^i = 1/\sqrt{N} \tilde{W}_t$$

# PROF. ORNSTEIN

## ONDERZOEKT DE

1930

## TOEVALS BEWEGING



$$dx_t = -\theta x_t dt + \sigma dW_t$$



DE STRAKKE HAND

Orinstein mural in Utrrecht

# McKean-Vlasov model

- Simple model: consider  

$$dX_t^i = \sum_{j=1}^N a(X_t^i - X_t^j)dt + \sigma dW_t^i$$
- What is the behaviour depending on  $a, \sigma, N$ ?
- Can be solved explicitly,  $Y_t^i = X_t^i - \frac{1}{N} \sum_i X_t^i$  is an Ornstein-Uhlenbeck process
- $dY_t^i = aNY_t^i dt + \sigma \left( dW_t^i - \frac{1}{N} \sum_i dW_t^i \right)$
- $dY_t^i = aNY_t^i dt + \sigma \sqrt{1 - 1/N} dB_t^i$ , consider  $a = c/N$
- For  $c < 0$ ,  $Y_t^i \rightarrow N(0, \tilde{\sigma}(N, c))$  in law as  $t \rightarrow +\infty$  and  $\frac{1}{N} \sum_i X_t^i = 1/\sqrt{N} \tilde{W}_t$



Ornstein mural in Utrecht

# Non-interacting ants

- 2D Active Brownian particles

- Particle  $i$  follows 
$$\begin{cases} dX_t^i &= \text{Pev}(\theta_t^i)dt + \sqrt{2\sigma_x}dW_t^i \\ d\theta_t^i &= \sqrt{2}dB_t^i \end{cases}$$